In the past two weeks of Sensing the World, I learned a lot of new skills and created a portfolio that will benefit my career. On the first day, we discussed the general programming approach and learned how to draw flowcharts. I was able to catch onto the meaning of each symbol quickly, and I understand how creating flowcharts is very helpful in grasping how the programming process works. Following, we created GitHub accounts and used Copilot to make a portfolio. I honestly think this is extremely valuable, and I am glad this course taught us how to make a portfolio. I will focus on completing my portfolio so I can use it in future internship/job applications. I was surprised with how fast AI can create a functional and professional-looking website. I knew this was a thing, but doing it myself really made me impressed with AI on a whole new level.

Next, we received our microcontrollers, the Arduino Nano ESP32, and installed the Arduino MicroPython application onto our computers. Our goal was to program a security alarm system. We created a flowchart to understand our objective and then programmed the microcontroller to run what we wanted. This intro to microcontrollers was very interesting because I have always been curious about how technology like an alarm system works, and it was intriguing to learn the basics of how the sensors are programmed. We viewed a zoomed-in version of the microcontroller and discussed the purpose of each pin. We also learned the symbols in MicroPython, specifically “#” representing a comment. I never knew that most code included comments, but now I realize why it is so important. I am able to look back at all of the code I created and immediately understand what each command is trying to do. One of the final things we did in the first week was creating a new GitHub repo for my FYRE assignments. Adding all of my programs and other projects in this course will be useful in the future to add to my resume and recall important steps in my learning journey.

For the second week of this section, we were out of the STEPS building and worked in the Keysight lab at Packard Hall. We learned about how to control circuits and then got right into hands-on creating a circuit with wires, resistors, and an LED bulb. We connected the breadboard to the Keysight power supply and explored what happens at different voltages. This circuit making made a lot of sense to me, and I enjoyed setting up the circuit in the exact right way. Then, we added in a microcontroller and used ChatGPT to generate MicroPython code that would run our objective on the breadboard. I feel like I am good with giving AI the right prompt, and I felt satisfied with it working correctly on the first try. It was helpful to get assistance on how to craft a specific prompt so ChatGPT writes the exact code I want. I like how we are learning the dramatic effect AI has on this part of engineering and engineering in general. Tasks such as this would take engineers hours and now can be done in seconds with a thoughtful prompt. It was also interesting to learn how the Arduino ground needs to be connected to the breadboard ground and how everything needs to be plugged into a specific pin to complete the circuit.

On the last day of this section, we worked with more complex attachments to the breadboards. We connected a micro servo motor to the breadboard, then used AI to program the motor to rotate 180 degrees and back. My group felt confident with making the right AI prompt, and it worked on the first try after we adjusted the breadboard components to make the right connection. Lastly, we connected a water sensor to the breadboard and used the Sinclair power supply to measure the voltage reading at different dry/wet conditions. We created an automated version of this with a MicroPython code and then saved all of our data to Google Sheets to display our results on a table and a graph. In the past, I used Google Sheets for

science classes in high school, but it was helpful to be reminded how the features work for a new purpose. Overall, I learned a lot about microcontrollers, programming/coding, and circuits. The past two weeks gave me a better sense of what I am truly passionate about, and I am looking forward to learning about more things like this and polishing my portfolio.